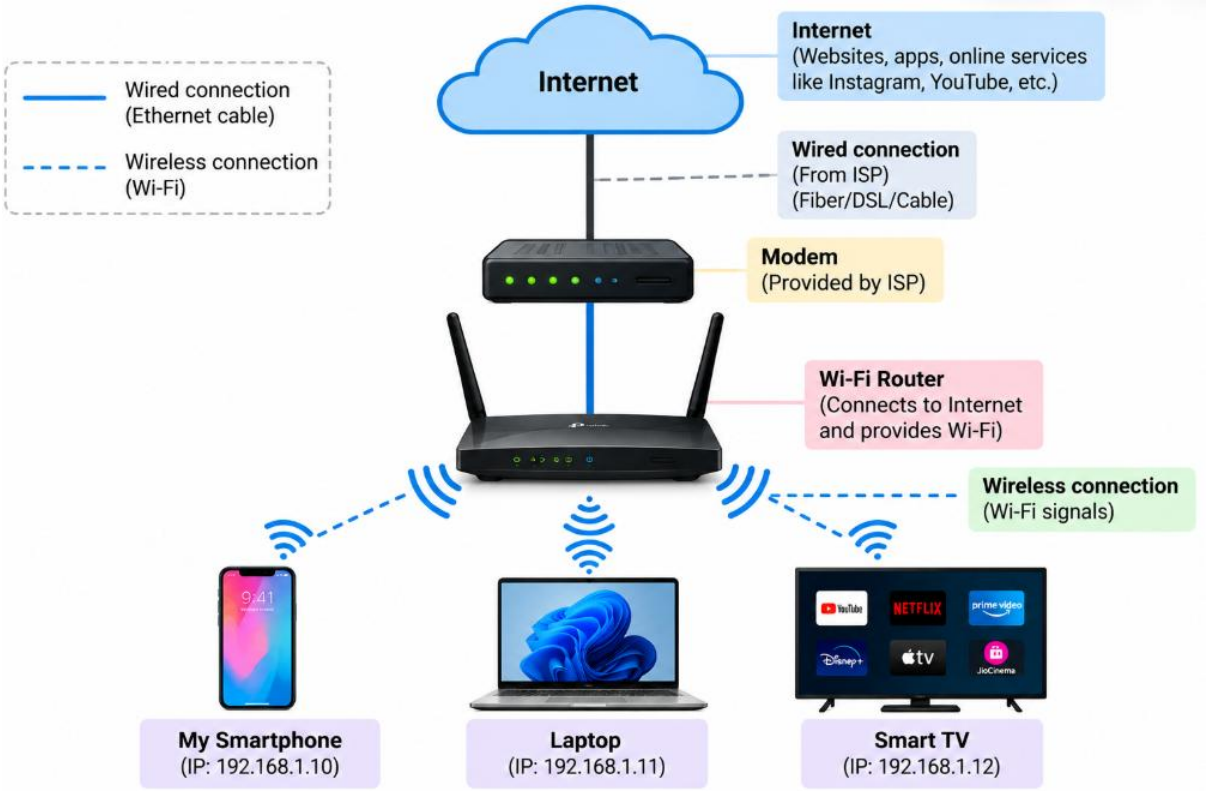
Wi-Fi Networks

1. Three Popular Wi-Fi Standards

Wi-Fi Standard	Frequency Band	Maximum Theoretical Speed	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Up to 600 Mbps	Supports both bands and provides good coverage.
802.11ac (Wi-Fi 5)	5 GHz	Several Gbps; up to about 6.9 Gbps	Offers higher speeds, especially for video streaming.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Up to about 9.6 Gbps	Handles many connected devices more efficiently.

Note: These are theoretical maximum speeds. Actual speeds depend on the router, device, distance, interference and network conditions.

2. Diagram: Smartphone Connecting to the Internet Through Wi-Fi



Explanation: The router connects to the Internet through a wired connection provided by the ISP. My smartphone connects wirelessly to the router using Wi-Fi. When I open an app such as Instagram, the router forwards my data to the Internet and delivers the response back to my phone.

3. Scan Available Wi-Fi Networks

You can scan nearby Wi-Fi networks using your laptop or smartphone and record their names, signal strength and security types.

Sample observation table

The following values are examples only. Replace them with the actual networks visible on your device.

No.	SSID (Network Name)	Signal Strength	Security Type
1	Home_WiFi	Excellent (90%)	WPA2-Personal
2	College_WiFi	Good (70%)	WPA2-Enterprise
3	Guest_Network	Weak (35%)	Open (No password)

How to scan on Windows

1. Click the Wi-Fi icon on the taskbar.
2. View the list of available Wi-Fi networks.
3. Record the SSIDs and approximate signal strength.
4. To inspect security details, open the network's properties or use Windows Command Prompt:

netsh wlan show networks mode=bssid

This command lists detected networks and can show authentication and encryption details.

Which network would I choose?

I would choose Home_WiFi because it has the strongest signal in this example and uses WPA2 security. If WPA3 is available and supported by my devices, I would prefer it. I would avoid connecting to an unknown open network when handling sensitive information.

4. What Is SSID Hiding?

SSID stands for Service Set Identifier, which is simply the name of a Wi-Fi network.

SSID hiding is a router setting that prevents the network name from being advertised normally in Wi-Fi network lists. Users who want to connect may need to enter the network name manually.

Advantage: It can reduce casual visibility of the network name to nearby users.

Disadvantage: It does not provide real security. Network discovery tools can still detect a hidden network, and authorized users may find connecting more difficult.

Conclusion: Hiding an SSID should not replace strong Wi-Fi security. Using WPA2-AES or WPA3 with a strong password is much more important.